Day 5 — Mid-module Check

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest the Day 1 baseline problem using the Nikhilam method, record the new time, and complete the formative quiz Part B that consolidates Days 1–4.

Materials

- The Day 1 baseline sheets (returned today)
- Fresh paper for the retest
- `quizzes/formative.md` Part B (printed)
- Stopwatch

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times (loud).
- Make-10 reflex: 10 quick partners.

Retest of Day 1 baseline (8 min)

- Hand back the Day 1 sheets so students can see their old time.
- Give a fresh sheet with the same problem: `10 000 − 4567 = ?`
- *"Solve it the Nikhilam way. Time yourself."*
- Record new time below the old time.
- **% improvement** = (Day 1 time − Day 5 time) / Day 1 time × 100. Compute on the spot.

Brief sharing (5 min)

- 3 volunteers share their % improvement.
- *"Some of you went from 30 sec to 8 sec. Some from 18 sec to 15 sec. Both are wins. The point is the trend."*

Formative quiz Part B (15 min)

- Hand out `quizzes/formative.md` Part B.
- 12 mixed problems: 3 complement-to-100, 4 complement-to-1000 / `1000 − N`, 3 `10 000 − N`, 1 four-digit padding case, 1 short answer ("state the sūtra in English and Sanskrit").
- Closed notebook. Aim to finish in 12 minutes.
- Collect for grading.

Diagnostic discussion (10 min)

- While the quiz is being graded (or based on a quick teacher scan), call out three common error types on the board:

    1. **Padding error** — student wrote `10000 - 89 = 9921` because they treated 89 as "8 9" not "0 0 8 9." Reteach: pad with leading zeros to match the base's digit count.
    2. **Last-digit error** — student wrote `9-7 = 2` for the units instead of `10-7 = 3`. Reteach: the LAST digit always takes from 10. Always.
    3. **Trailing-zero handling** — `1000 - 70`: digit-wise gives `9-0, 9-7, 10-0 = 9, 2, 10`. The 10 carries: `9, 3, 0` → 930. Verify: 70 + 930 = 1000. ✓. Acknowledge this is a corner case; offer the easier "see 70 as 070, complement directly" framing.
- Take questions.

Wrap-up (2 min)

- Project choice form due tomorrow. Three options: shopkeeper chart, teaching video, two's-complement poster. (See `assessments/project-brief.md`.)
- Preview Day 6: *"Why does this work? Tomorrow we prove it — with algebra."*

Student-facing content

Mid-module Check

Day 1 time: ____ sec → Day 5 time: ____ sec → % change: ____

If you improved less than expected, the most common reason is one of three things:

1. Padding error — you forgot leading zeros.
2. Last-digit error — you used "from 9" for the last digit instead of "from 10."
3. You're still translating mentally between "borrow" and "complement."

Don't panic. Days 6–9 fix all three.

Homework

- Choose your project format (shopkeeper chart / video / poster) and write a 3-sentence plan.
- 5 mixed Nikhilam subtractions: `1000 - 358`, `10 000 - 6427`, `1000 - 72`, `100 000 - 12 345`, `10 000 - 9`. (Last one is `9991`; check the padding!)

Differentiation

- **Tier up:** Try mixed-base, mixed-method speed test: 6 problems in 60 seconds. Use Nikhilam where helpful, standard where not.
- **Tier down:** Re-do the Day 4 worksheet. Don't take the quiz cold.

Teacher notes

- The retest is the emotional pivot of the module. Most students will see a real improvement; honor it without overselling. (“You’re 50% faster — and you’re still getting faster.”)
- Use the formative quiz to triage: anyone scoring below 60% gets a 10-min pull-aside on Day 7 before the two’s-complement lesson. Don’t let them carry a foundational gap into the harder material.
- Distinguish: this is a *formative* check, not summative. No grade weight. Day 10 is summative.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).